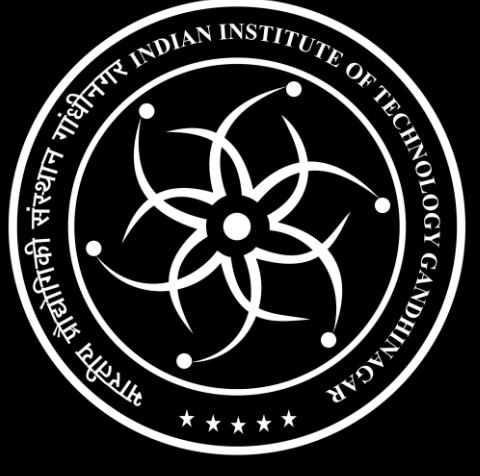


Environmental Quality Monitoring System Using Low-Cost Sensors for Urban Air Quality Monitoring and Multi-Scale Forecasting



ENERGY, ENVIRONMENT,
AND EXPOSURE LABORATORY
INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR

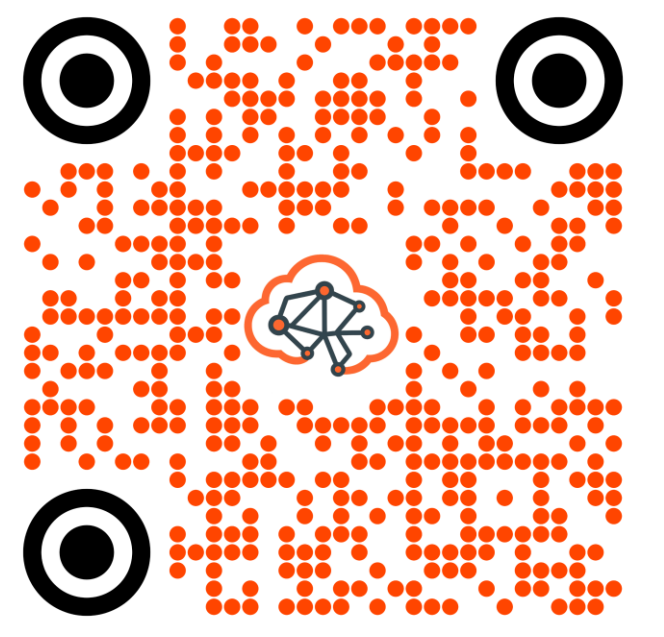
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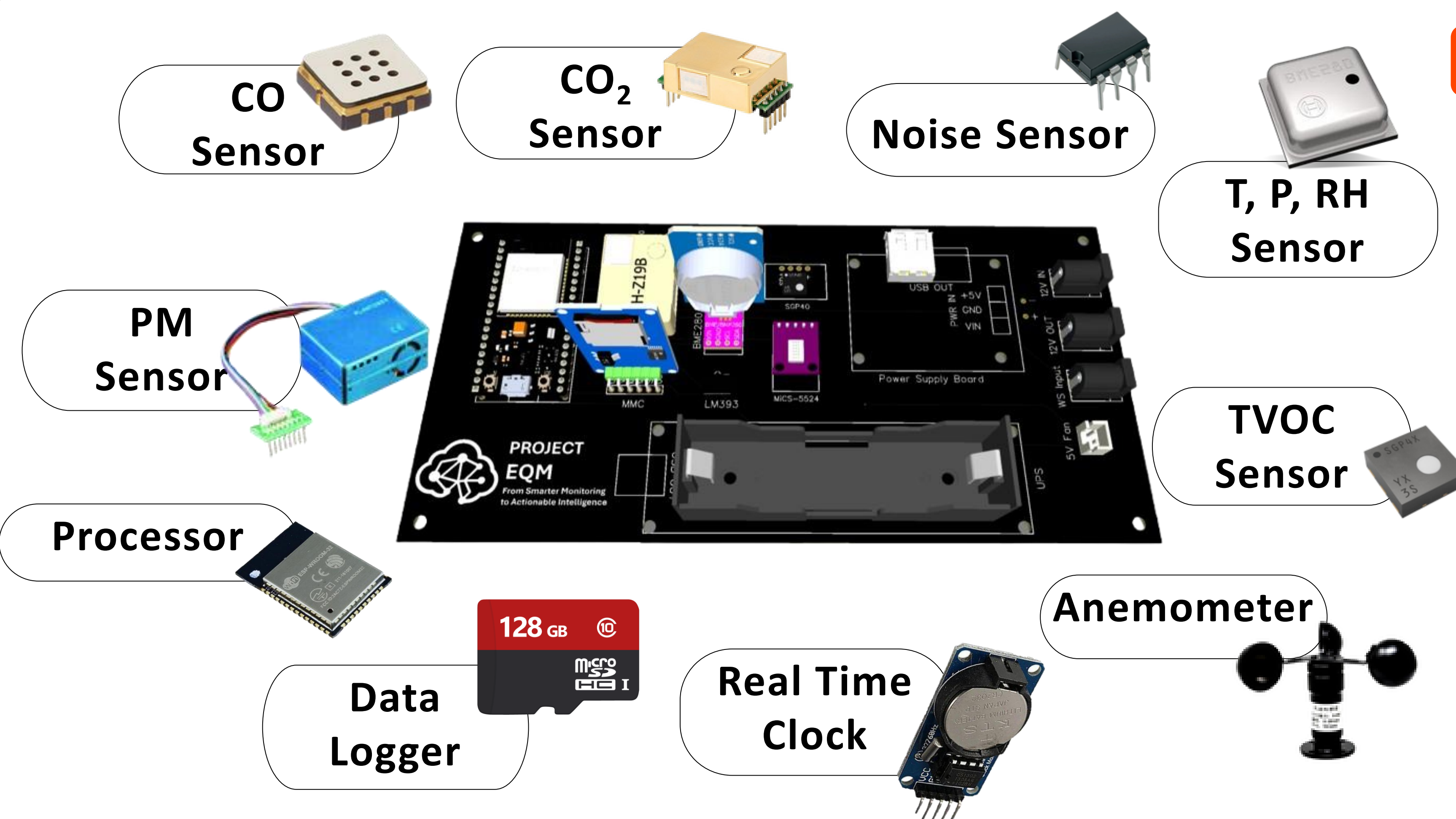
PROJECT
EQM
From Smarter Monitoring
to Actionable
Intelligence



- Focuses on the design and deployment of an **integrated low-cost sensor network** for comprehensive monitoring of PM, CO, CO₂, noise, temperature, humidity, wind, and TVOCs.
- Facilitates **city-wide, real-time environmental data acquisition** through a centralized server, enabling seamless monitoring across urban landscapes.
- Utilizes a **hybrid CNN-BiLSTM framework** for anomaly detection, spatio-temporal analysis, and multi-scale forecasting (24-hour, 7-day, 1-month) of air quality dynamics.
- Translates complex environmental data into **actionable insights**, embodying the motto: **FROM SMARTER MONITORING TO ACTIONABLE INTELLIGENCE**.



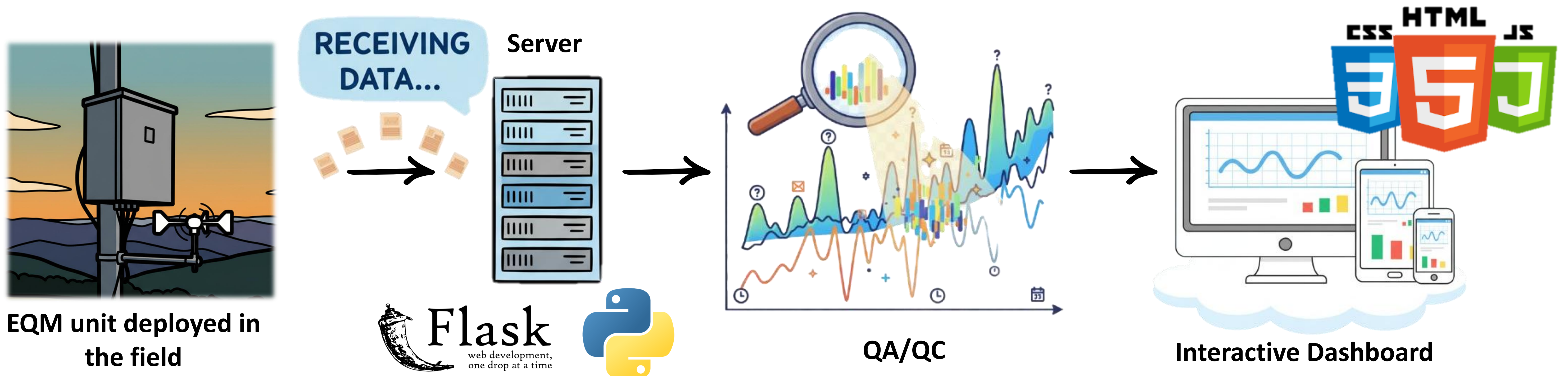
Scan for
Dashboard



DIFFERENT COMPONENTS OF EQM

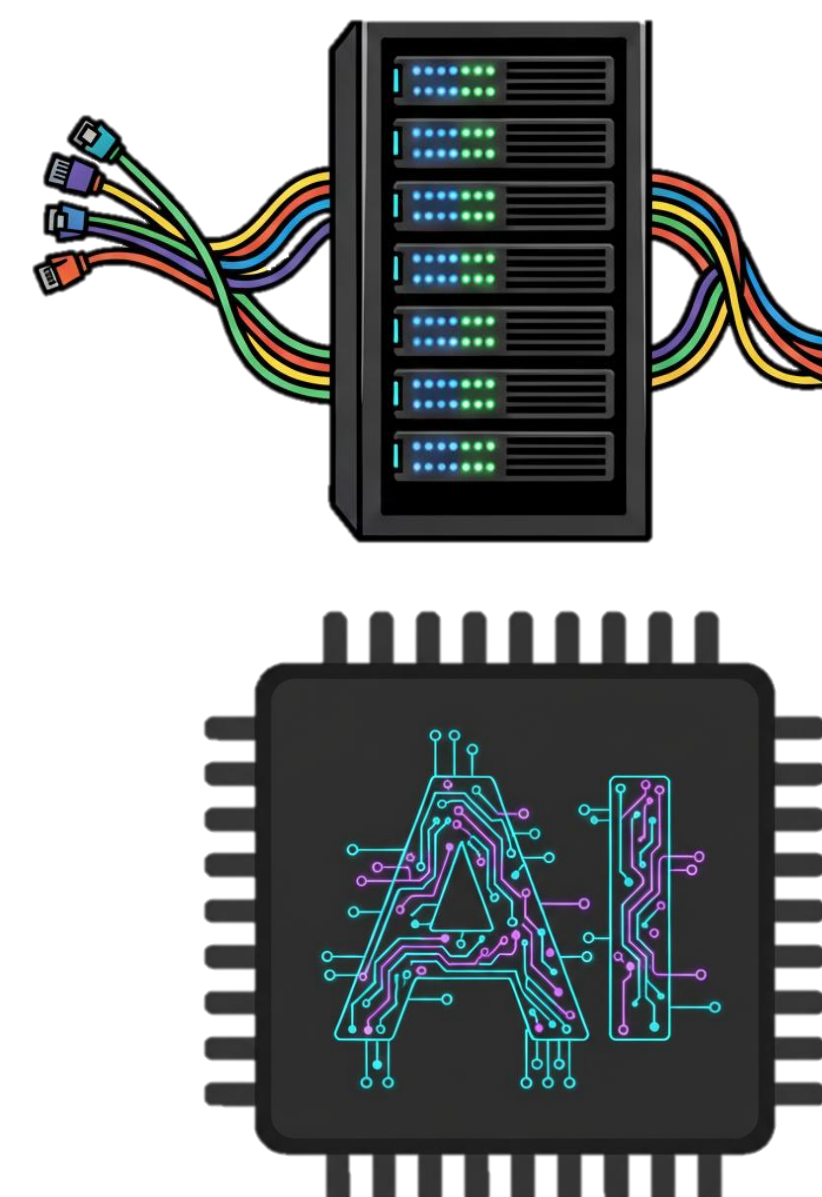
- PM Sensor** – Measures PM₁, PM_{2.5}, PM₁₀ concentrations.
- CO₂ Sensor** – Monitors carbon dioxide for air quality and ventilation.
- CO Sensor** – Detects carbon monoxide levels.
- TVOC Sensor** – Detects total volatile organic compounds.
- T, P, RH Sensor** – Measures temperature, pressure, and relative humidity.
- Anemometer** – Measures wind speed and direction.
- Noise Sensor** – Records ambient noise for pollution analysis.
- Processor** – Integrates sensors, processes data, and manages communication.
- Real Time Clock** – Provides accurate timestamps.
- Data Logger** – Stores large sensor datasets.

NETWORK ARCHITECTURE



Server

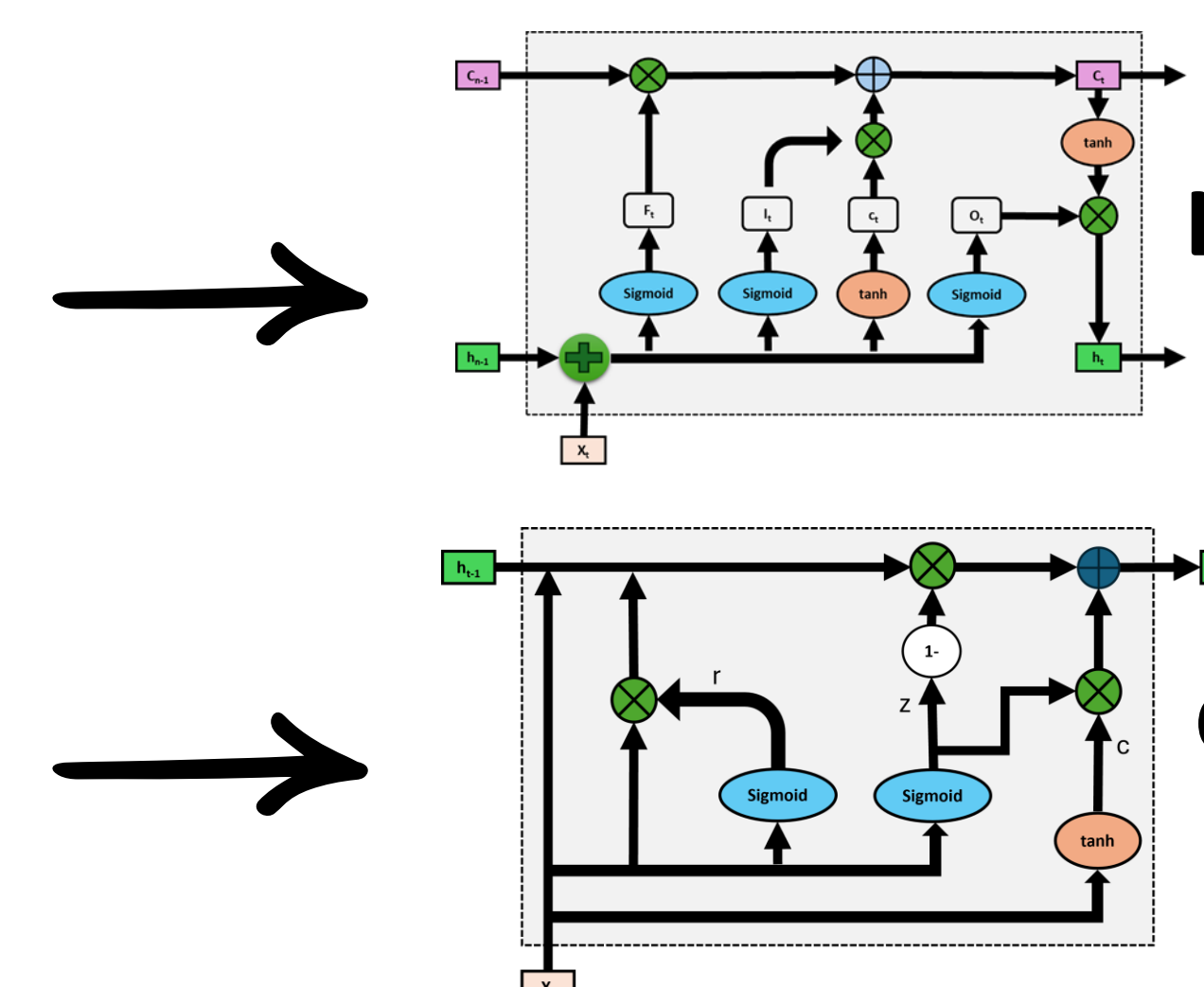
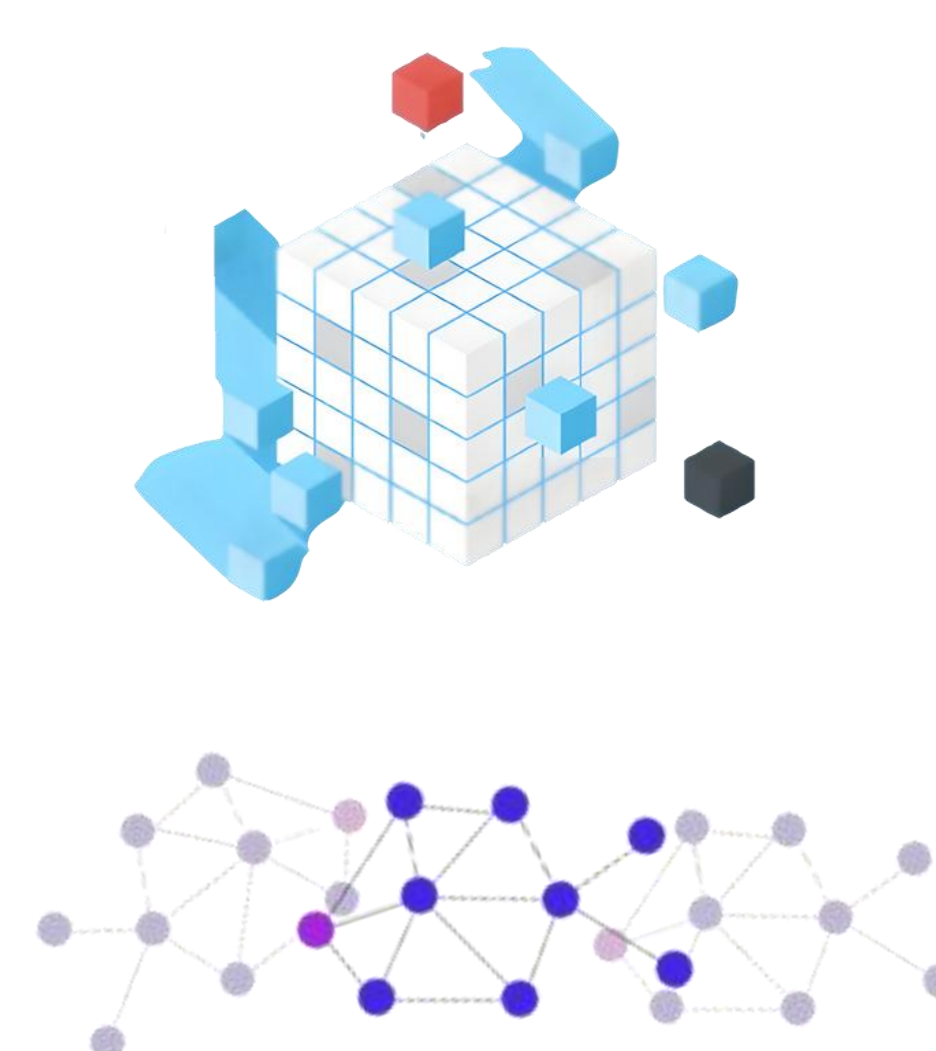
- CNN-BiLSTM / GNN-GRU:** Capture spatial + temporal patterns for prediction.
- Spatiotemporal Processing:** Combine location & time data for analysis.
- Anomaly Detection:** Spot sudden spikes, drops, or sensor errors.



CNN

GNN

DATA ANALYSIS



OUTPUT

DEPLOYMENTS

- Initial trial deployment planned at IITGN campus for testing and validation.
- Large-scale deployment of 50–60 EQM units across Ahmedabad for city-wide, real-time monitoring.
- Strengthens environmental data collection by providing continuous and distributed measurements.
- Enables monitoring of air quality, noise levels, and other key environmental parameters for better urban management.

Project Collaborator:
Prof. Sushobhan Sen, IITGN